A Fault Diagnostic Algorithm of General Biswapped Networks *

Jheng-Cheng Chen and Chang-Hsiung Tsai†
Department of Computer Science and Information Engineering
National Dong Hwa University
Hualien, Taiwan 970, R.O.C.

Abstract

Processor fault identification in a multiprocessor system plays an important role in reliable computing. The process of identifying faulty processors is called the diagnosis of the system. Several diagnostic models have been proposed, the most popular being the PMC (Preparata, Metze and Chen) diagnostic model proposed in 1967. General biswapped networks were proposed in 2007 as interconnection networks amenable to implementation using a mix of electronic and optical communication links. The model is inspired by and extended from biswapped networks and OTIS networks. In this paper, we propose a pessimistic diagnosis algorithm for general biswapped networks. Furthermore, the algorithm can run in $O(t_1N)$ *time where* N *denotes the total number of nodes in the network.*

1 Introduction

Multiprocessor systems are vulnerable to faults, and fault-tolerant computing has become very important for systems with several processors linked by an interconnection network. To maintain the reliability of a system, any faulty processor should be replaced with a fault-free one. The method of finding faulty processors is called fault diagnosis of the system. The degree of diagnosability of the system denotes the maximum number of faulty processors that can be accurately identified in the system.

System-level diagnosis is an important technique for improving system reliability and availability. A graph-theoretical model of system-level fault diagnosis termed the PMC model was proposed in a classic paper by Preparata, Metze and Chien [12]. In this model, every processor in a system performs tests on its neighbors utilizing communication links between them. When one processor tests another, the tester declares the tested processor to be fault-free or faulty depending on the test response; the result is always accurate if the tester is fault-free, but if the tester is faulty, the result is unreliable. According to the traditional precise diagnosis strategy, it is demanded that all processors be identified correctly, specifically that all fault-free processors are identified as 'fault-free' and all faulty processors are identified as 'faulty' [12]. The pessimistic diagnosis strategy proposed by Kavianpour and Friedman [6] is a process to diagnose faults that allows all faulty processors can be isolated within a set having at most one fault-free processor. In order to identify faults, a number of tests must be performed among processors and the collection of all test results is referred to as a syndrome.

Under the PMC model with a pessimistic strategy, the diagnosabilities of some well-known multicomputer systems for example hypercubes [7], enhanced hypercubes [14], Möbius cubes [4], and star graphs [8] have been determined in the last few decade. Recently, Chang et al. [1] investigated the pessimistic diagnosabilities of regular networks and provided sufficient conditions for computing the pessimistic diagnosabilities of regular networks.

General biswapped networks were introduced in [5] as interconnection networks amenable to implementation using a mix of electronic and optical communication links. The model is inspired by and extended from Biswapped Networks and OTIS networks [11, 15, 16, 17]. A general biswapped network (GBSN) $Gbsw(G, H)$*, where* G *is the base graph and* H *is the network graph, is composed of* m *copies of some* n*-node basis network* G *and* n *copies of some* m*-node basis network* H*. Such a network uses a simple rule for connectivity that ensures its semi-regularity, modularity, fault tolerance, and algorithmic efficiency. In particular, the BSNs are special cases in which the two ba-*

*This work was supported in part by the National Science Council of the Republic of China under Contract NSC 101-2115-M-259-006-.

†Correspondence to: e-mail: chtsai@mail.ndhu.edu.tw.

sis networks are the same. The Exchanged Hypercube is another example of GBSNs, i.e., $EH(s,t)$ is isomorphic to $Gbsw(H(s),H(t))$.

The objective of this paper is to study the diagnosability of general biswapped networks under the PMC model with the the pessimistic diagnosis strategy. Since $Gbsw(H,G)$ is t-diagnosable and t_1/t_1-diagnosable under the precise and pessimistic strategies, respectively, where t is minimum node degree of $Gbsw(H,G)$ and t_1 is the minimum number of neighbors of any pair of nodes, a pessimistic diagnosis algorithm for general biswapped networks is proposed in this paper and the algorithm can run in $O(t_1N)$ time where N denotes the total number of nodes in the network.

The rest of this paper is organized as follows: Section 2 provides graph-theoretic terminology and fundamental properties. Section 3 is focused on a pessimistic diagnosis algorithm for general biswapped networks, including a formal description, correctness proof, and time complexity analysis. Finally, our conclusions are given in Section 4.

2 Definitions and Basic Properties

The topology of a multicomputer system is often represented by a graph $G=(V,E)$, where each of the vertices $u \in V$ denotes a processor and each edge $(u,v) \in E$ denotes a bidirectional communication link between two processors u and v. Hereafter, we speak of vertex and edge instead of processor and communication link. Throughout this paper the three terms - multicomputer, system and graph - will be used interchangeably. In this paper, we focus on an undirected graph without loops and follow [10] for the graph definitions and notations. $G=(V,E)$ is a graph if V is a finite set and E is a subset of $\{(u,v) \mid (u,v)$ is an unordered pair of $V\}$. We say that V is the vertex set and E is the edge set. Two vertices u and v are adjacent if $(u,v) \in E$; the vertex u is called a neighbor of v, and vice versa. The degree of vertex u is denoted by $deg(u)$ and the degree of graph G is denoted by $\delta(G) = min_{u \in V(G)} deg(u)$. A graph G' is a subgraph of a graph G if $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$.

He et al. [5] proposed a class of graphs, called a general biswapped graphs as following: Let G be any undirected graph with the vertex set $V(G) = \{g_1, g_2, \ldots, g_n\}$ and the edge set $E(G)$. And let H be any graph with the vertex set $V(H) = \{h_1, h_2, \ldots, h_m\}$ and the edge set $E(H)$. The General Biswapped Network $Gbsw(G,H)$ is a graph composed of m copies of G, $\{G_1, G_2, \ldots, G_m\}$, and n copies of H, $\{H_1, H_2, \ldots, H_n\}$, with additional external links in between. The vertex and edge sets of $Gbsw(G,H)$ is specified as: $V(Gbsw(G,H)) = \{g_{ij} \mid g_{ij} \in V(G_i), 1 \le i \le m, 1 \le j \le n\} \cup \{h_{ji} \mid h_{ji} \in V(H_j), 1 \le i \le m, 1 \le j \le n\}$ and $E(Gbsw(G,H)) = \{(g_{ir}, g_{is}) \mid (g_r, g_s) \in E(G), 1 \le i \le m, 1 \le r,s \le n\} \cup \{(h_{jr}, h_{js}) \mid (h_r, h_s) \in E(H), 1 \le j \le n, 1 \le r,s \le m\} \cup \{(g_{ij}, h_{ji}) \mid g_{ij} \in V(G_i), h_{ji} \in V(H_j), 1 \le i \le m, 1 \le j \le n\}$.

Intuitively, the definition postulates two parts: m clusters of G (part 0) and n clusters of H (part 1) with inter-cluster links. Denote g_{ij} as $\langle 0,i,j \rangle$ and h_{ji} as $\langle 1,j,i \rangle$, representing $\langle$ part#, cluster#, node# $\rangle$, for convenience. The edge set of it is composed of three parts: intra-cluster edges in G_i (part 0), intra-cluster edges in H_j (part 1), and inter-cluster or swapping edges between two parts. From the above definition, it is clearly that G and H plays a symmetric role in the composition of $Gbsw(G,H)$. Therefore, $Gbsw(G,H)$ is isomorphic to $Gbsw(H,G)$. For example, Fig. 1 shows two specific examples of GBSNs, in which (a) is $Gbsw(K_2,C_3)$ and (b) is $Gbsw(C_3,C_4)$.

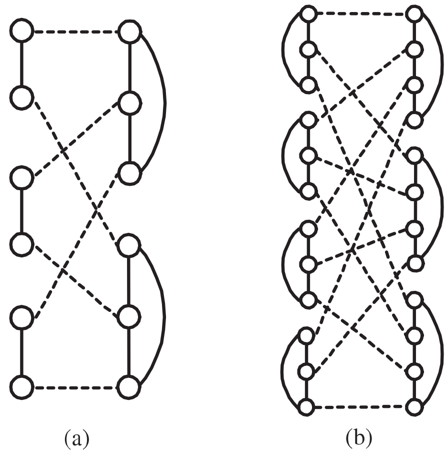

Figure 1: Examples of GBSNs. $(a)Gbsw(K_2,C_3)$ and $(b)Gbsw(C_3,C_4)$.

By definition of a general biswapped network, the following lemma is obvious obtained.

Lemma 1 *$\delta(Gbsw(G,H)) = min\{\delta(G), \delta(H)\} + 1$.*

Under the PMC model, a self-diagnosable system G was often modeled by a digraph in which an *arc direct from vertex u to vertex v means that u can test v. In this situation, u is a tester and v is a tested vertex. We use a graph $G = (V, E)$ to represent a self-diagnosable system.*

Definition 1 *Under the PMC model, a syndrome σ for system G is defined as follows: For any two adjacent vertices u and v,*

$$\sigma(u, v) = \begin{cases} 0, & \text{if } v \text{ is tested by } u \text{ to be fault-free;} \\ 1, & \text{if } v \text{ is tested by } u \text{ to be faulty.} \end{cases}$$

We have the following definition of diagnosability under a precise strategy.

Definition 2 *A graph G is t-diagnosable if all the faulty nodes can be identified without replacement, provided that the number of faults presented does not exceed t.*

The pessimistic diagnosability measure of the diagnostic capability of a graph is defined as follows:

Definition 3 *[6] A graph G is pessimistically t_1/t_1-diagnosable if all the faulty vertices can be isolated to within a set of at most t_1 vertices, provided that the number of faulty vertices at any give time is bounded from above by t_1. The pessimistic diagnosability of G is the maximum number t_1 such that G is pessimistically t_1/t_1-diagnosable.*

Lemma 2 *[9] The graph G is t-diagnosable if and only if, for each vertex set $S \subset V(G)$ with $|S| = p$, $0 \le p \le t-1$, every connected component C of $G - S$ satisfies $|V(C)| \ge 2(t-p)+1$.*

Theorem 1 *[2] For any two graphs G and H with $\delta(G) \ge 1$ and $\delta(H) \ge 1$, a general biswapped graph $Gbsw(G, H)$ is t_1/t_1-diagnosable where $t_1 = min_{u,v \in V}\{|N(\{u, v\})|\}$.*

3 A pessimistic diagnostic algorithm for general biswapped networks

In [13], a structure called the twinned star is proposed for helping to identify the status of one vertex in a t_1/t_1-diagnosable graph under the PMC model.

Definition 4 *[13] Letting $G = (V_G, E_G)$ be a graph, $(u, v) \in E_G$ be an edge, and k be an integer, a twinned star of order k at vertices u and v is defined to be a subgraph of G, denoted by $TS_G(u, v; k)$, where $V(TS_G(u, v; k)) = \{u, v\} \cup \{v_i, x_i \mid 1 \le i \le r\} \cup \{u_i, y_i \mid 1 \le i \le k-r\}$ and $E(TS_G(u, v; k)) = \{(u, v)\} \cup \{(v, v_i), (v_i, x_i) \mid 1 \le i \le r\} \cup \{(u, u_i), (u_i, y_i) \mid 1 \le i \le k-r\}$. (See Fig. 2).*

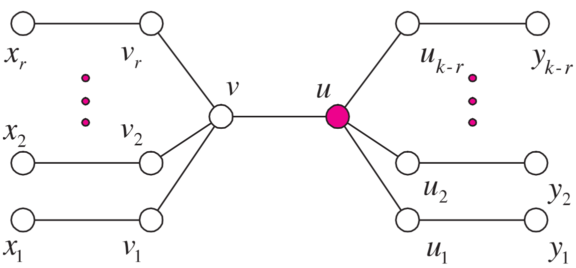

Figure 2: A twinned star $TS_G(u, v; k)$ at vertices u and v consists of $2k+2$ vertices and $2k+1$ edges.

Let σ be a syndrome of G and $n_{\alpha_1\alpha_2}(u) = |\{1 \le j \le k-r \mid (\sigma(y_j, u_j), \sigma(u_j, u)) = (\alpha_1, \alpha_2)\}|$ and $n_{\alpha_1\alpha_2}(v) = |\{1 \le j \le r \mid (\sigma(x_j, v_j), \sigma(v_j, v)) = (\alpha_1, \alpha_2)\}|$, where $\alpha_1, \alpha_2 \in \{0, 1\}$. In practice, a simple decision technique to identify the status of at least one of the two vertices u and v in $TS_G(u, v; k)$ was proposed as follows.

Theorem 2 *[13] Let $G = (V, E)$ be a t_1/t_1-diagnosable graph and $TS_G(u, v; t_1)$ be a twinned star of order t_1 at the vertex u and v. Then, one of following statements holds.*

(1) *Let $\sigma(v, u) = 0$. If $n_{00}(u) + n_{00}(v) < n_{01}(u) + n_{01}(v)$, then v must be faulty; otherwise, u must be fault-free.*

(2) *Let $\sigma(v, u) = 1$. If $n_{00}(u) + n_{01}(v) < n_{01}(u) + n_{00}(v)$, then u must be faulty; otherwise, v must be faulty.*

The twinned star can be constructed in many multiprocessor systems and interconnection networks, for example, hypercubes, crossed cubes, twisted cubes [13], etc. Hereafter, we describe how to construct a twinned star, namely $TS(u, v; t_1)$, in $Gbsw(G, H)$ for any given swapping edge (u, v) where the pessimistic diagnosability of $Gbsw(G, H)$ is t_1.

Let (u, v) be a swapping edge in $Gbsw(G, H)$, i.e., $u \in V_{G_i}$ and $v \in V_{H_j}$ for some clusters G_i and H_j. Without loss of generality, assume $u = g_{ij}$ and $v = h_{ji}$. Then a twinned star of order t_1 at vertices g_{ij} and h_{ji} in $Gbsw(G, H)$ can be formed by the graph $TS(g_{ij}, h_{ji}; t_1)$, whose vertex set and edge set are $V_{TS(g_{ij}, h_{ji}; t_1)} = \{g_{ij}, h_{ji}\} \cup$

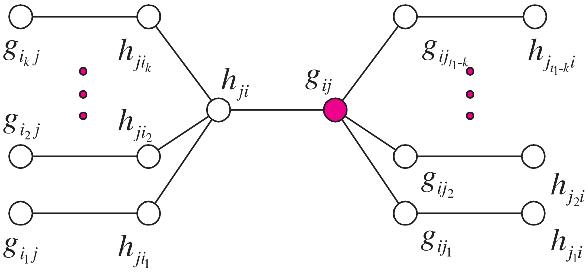

Figure 3: A twinned star $TS(u, v; t_1)$ at vertices g_{ij} and h_{ji} where (g_{ij}, h_{ji}) is a swapping edge in $Gbsw(G, H)$.

$\{g_{ij_s}, h_{j_s i} \mid 1 \le s \le t_1 - k\} \cup \{h_{ji_r}, g_{i_r j} \mid 1 \le r \le k\}$ and $E_{TS(g_{ij}, h_{ji}; t_1)} = \{(g_{ij}, h_{ji})\} \cup \{(g_{ij}, g_{ij_s}), (g_{ij_s}, h_{j_s i}) \mid 1 \le s \le t_1 - k\} \cup \{(h_{ji}, h_{ji_r}), (h_{ji_r}, g_{i_r j}) \mid 1 \le r \le k\}$, respectively. See Fig. 3 for illustration. Consequently, the twinned star structure of order t_1 in $Gbsw(G, H)$ can be constructed in time $O(t_1)$.

*Next, using the twinned star structure we can design an efficient algorithm, namely Diagnose-the-Given-Swapping-Edge (***DGSE***, for short), to determine the status of one of the end-vertices in a given swapping edge (g_{ij}, h_{ji}) in $Gbsw(G, H)$. See Algorithm 1.*

*Finally, we propose an efficient algorithm, namely Pessimistic-Diagnose-General-Biswapped-Network (***PDGBSN***, for short), to diagnose $Gbsw(G, H)$ under the PMC model with a pessimistic strategy. See Algorithm 2. A t_1/t_1-diagnosable graph G is said to be pessimistic diagnosable if there is an algorithm to identify all faulty nodes of G within a set containing at most one fault-free node. Subsequently, we take into account the correctness and the time complexity of algorithm* **PDGBSN***. An auxiliary lemma is needed as follows.*

Let U be a set of vertices in $Gbsw(G, H)$. The notation $U(G_i)$ (resp., $U(H_j)$) denotes $U \cap V_{G_i}$ (resp., $U \cap V_{H_j}$) for every cluster G_i (resp., H_j) of $Gbsw(G, H)$. Two vertex sets W and Z are connected if there is an edge between them; otherwise, they are disconnected. The vertex set U in $Gbsw(G, H)$ is said to be dividable by clusters if any two different subsets $U(X)$ and $U(Y)$ of U are disconnected where X, Y are two different clusters, copies of G and/or H, in $Gbsw(G, H)$.

Lemma 3 *Let U be a dividable vertex set in $Gbsw(G, H)$ that includes two vertices at least. Then, $|N_{Gbsw(G,H)}(U)| \ge t_1$ where $t_1 = min_{u,v \in V} |N_{Gbsw(G,H)}(\{u, v\})|$.*

Algorithm 1 DGSE($Gbsw(G, H)$, (g_{ij}, h_{ji}))

$t_1 \leftarrow min_{u,v \in V} |N_{Gbsw(G,H)}(\{u, v\})|$.
Construct a twinned star $TS(g_{ij}, h_{ji}; t_1)$ as illustrated in Fig. 3.
$n_{00}(g_{ij}) \leftarrow |\{1 \le s \le t_1 - k \mid (\sigma(h_{j_s i}, g_{ij_s}), \sigma(g_{ij_s}, g_{ij})) = (0, 0)\}|$
$n_{01}(g_{ij}) \leftarrow |\{1 \le s \le t_1 - k \mid (\sigma(h_{j_s i}, g_{ij_s}), \sigma(g_{ij_s}, g_{ij})) = (0, 1)\}|$
$n_{00}(h_{ji}) \leftarrow |\{1 \le r \le k \mid (\sigma(g_{i_r j}, h_{ji_r}), \sigma(h_{ji_r}, h_{ji})) = (0, 0)\}|$
$n_{01}(h_{ji}) \leftarrow |\{1 \le r \le k \mid (\sigma(g_{i_r j}, h_{ji_r}), \sigma(h_{ji_r}, h_{ji})) = (0, 1)\}|$
if $(\sigma(h_{ji}, g_{ij}) = 0)$ **then**
  if $((n_{00}(g_{ij}) + n_{00}(h_{ji})) < (n_{01}(g_{ij}) + n_{01}(h_{ji})))$ **then**
    return h_{ji} faulty
  else
    return g_{ij} fault-free
  end if
else
  if $((n_{00}(g_{ij}) + n_{01}(h_{ji})) < (n_{01}(g_{ij}) + n_{00}(h_{ji})))$ **then**
    return g_{ji} faulty
  else
    return h_{ij} faulty
  end if
end if

Proof. *Let $U = \{U(X_1), U(X_2), \ldots, U(X_k)\}$ where X_i is a cluster of $Gbsw(G, H)$ and $X_i \ne X_j$ for $i \ne j$, i.e., X_i is a copy of G or a copy of H. Thus, $U(X_i)$ and $U(X_j)$ are disconnected if $X_i \ne X_j$. For simplicity, let $\overline{X}$ denote the graph obtained by $Gbsw(G, H)$ deleting all clusters X_i for $i = 1, \ldots, k$ and also let X denote the graph induced by $\bigcup_{i=1}^{k} V_{X_i}$. Therefore, $N_{Gbsw(G,H)}(U) = N_X(U) \cup N_{\overline{X}}(U)$ and $|N_{\overline{X}}(U)| = |U|$, where $N_X(U) \cap N_{\overline{X}}(U) = \emptyset$.*

Let u and v be two different vertices in U. Obviously, $N_{Gbsw(G,H)}(\{u, v\}) = N_X(\{u, v\}) \cup N_{\overline{X}}(\{u, v\})$ and $|N_{\overline{X}}(\{u, v\})| = 2$. Therefore, $|N_X(\{u, v\})| \ge t_1 - 2$ since $|N_{Gbsw(G,H)}(\{u, v\})| \ge t_1$. It is obvious that $|N_{X-U}(\{u, v\})| \ge |N_X(\{u, v\})| - |U| + 2 \ge t_1 - |U|$. Since $u, v \in U$, $N_{X-U}(\{u, v\}) \subseteq N_X(U)$. Hence $|N_X(U)| \ge t_1 - |U|$. As a result, $|N_{Gbase(G,H)}(U)| \ge t_1$. □

Now we are ready to analyze the complexity and correctness of the algorithm **PDGBSN***.*

Theorem 3 *A general biswapped graph $Gbsw(G, H)$ with N vertices containing at most t_1 faulty vertices is pessimistic diagnosable by the algorithm* **PDGBSN** *in time*

Algorithm 2 PDGBSN($Gbsw(G,H)$, σ)

Require: A faulty general biswapped graph $Gbsw(G,H)$ including at most t_1 faulty vertices and a syndrome σ.

Ensure: A set including all faulty vertices and containing at most one fault-free vertex.

1: $t_1 \leftarrow min_{u,v \in V}|N_{Gbsw(G,H)}(\{u,v\})|$.
2: $U \leftarrow V$, $F \leftarrow \emptyset$, $M \leftarrow \emptyset$
3: **for** each swapping edge (g_{ij}, h_{ji}) in $Gbsw(G,H)$ **do**
4:   call algorithm **DGSE**$(Gbsw(G,H),(g_{ij},h_{ji}))$
5:   **if** (g_{ij} is faulty) **then**
6:     $F \leftarrow F \cup \{g_{ij}\}$ and $U \leftarrow U - \{g_{ij}\}$
7:     **if** $(\sigma(h_{ji}, g_{ij}) = 0)$ **then**
8:       $F \leftarrow F \cup \{h_{ji}\}$ and $U \leftarrow U - \{h_{ji}\}$
9:     **end if**
10:   **end if**
11:   **if** (h_{ji} is fault-free) **then**
12:     $M \leftarrow M \cup \{h_{ji}\}$ and $U \leftarrow U - \{h_{ji}\}$
13:     **if** $(\sigma(h_{ji}, g_{ij}) = 1)$ **then**
14:       $F \leftarrow F \cup \{g_{ij}\}$ and $U \leftarrow U - \{g_{ij}\}$
15:     **else**
16:       $M \leftarrow M \cup \{g_{ij}\}$ and $U \leftarrow U - \{g_{ij}\}$
17:     **end if**
18:   **else**
19:     $F \leftarrow F \cup \{h_{ji}\}$ and $U \leftarrow U - \{h_{ji}\}$
20:     **if** $(\sigma(g_{ij}, h_{ji}) = 0)$ **then**
21:       $F \leftarrow F \cup \{g_{ij}\}$ and $U \leftarrow U - \{g_{ij}\}$
22:     **end if**
23:   **end if**
24:   **if** $(|F| = t_1)$ **then**
25:     $M \leftarrow M \cup U$
26:     **return** F
27:   **end if**
28: **end for**
29: **while** (there is an edge (u,v) between U and M where $u \in U$ and $v \in M$) **do**
30:   **if** $(\sigma(v,u) = 0)$ **then**
31:     $M \leftarrow M \cup \{u\}$ and $U \leftarrow U - \{u\}$
32:   **else**
33:     $F \leftarrow F \cup \{u\}$ and $U \leftarrow U - \{u\}$
34:   **end if**
35:   **if** $(|F| = t_1)$ **then**
36:     $M \leftarrow M \cup U$
37:     **return** F
38:   **end if**
39: **end while**
40: **return** $F \cup U$

$O(t_1N)$, where $\delta(G) \geq 1$, $\delta(H) \geq 1$, and $t_1 = min_{u,v \in V}|N_{Gbsw(G,H)}(\{u,v\})|$.

Proof. *Since every node of $Gbsw(G,H)$ is exactly incident to a swapping edge, there are $\frac{N}{2}$ different swapping edges. Since every swapping edge (u,v) was examined once by the twinned star $TS(u,v;t_1)$ in the algorithm* **DGSE**, *at least one of the end-vertices of an edge (u,v) should be identified as being faulty or fault-free by the algorithm* **DGSE** *taking at most $O(t_1)$ time. Therefore, $|U| \leq |F|$ when statements 3-28 of the algorithm* **PDGBSN** *are completed. And also, performing statements 3-28 needs at most $O(t_1N)$ time.*

After completing statements 3-28 of the algorithm, the diagnosis is complete and all faulty nodes are identified and collected to F if $|F| = t_1$. If $|F| \leq t_1 - 1$, the diagnostic process is going to carry out statements 29 to 40. Thus, $|U| \leq t_1 - 1$ and U is a dividable vertex set. By Lemma 3, $|N_{Gbase(G,H)(U)}| \geq t_1$. Since the process is stopped when $|F| = t_1$, there is at least one vertex in M connecting to one vertex in U if $|U| \geq 2$ and $|F| \leq t_1 - 1$. Therefore, $|U| \leq 1$ when statements 29-40 are completed and $|F| \leq t_1 - 1$. It is obvious that statements 29-40 need at most $O(t_1^2)$ time. Finally, the algorithm **PDGBSN** *isolates all faulty nodes to within a set $F \cup U$ of size at most t_1 and at most one vertex's status is unknown. The total time needed by this algorithm is $O(t_1N)$.* □

4 Conclusions

Optically interconnected technologies can offer several advantages over electronic implementations. General biswapped networks were proposed as interconnection networks amenable to implementation using a mix of electronic and optical communication links. This model is inspired by and extended from biswapped networks and OTIS networks, and uses a simple rule for connectivity, called biswapping strategy, that ensures its semi-regularity, modularity, fault tolerance, and algorithmic efficiency.

In this paper, we propose a pessimistic diagnosis algorithm for general biswapped networks and the algorithm can run in $O(t_1N)$ time where N denotes the total number of nodes in the network. The method developed in this paper should be expandable to design efficient diagnosis algorithms on other hierarchical interconnection networks.

References

[1] G. Y. Chang, G. J. Chang, and G. H. Chen "Diagnosabilities of regular networks", IEEE Trans. Parallel and Distributed Systems, vol. 16, no. 4, pp. 314-323, 2005.

[2] J. C. Chen and C. H. Tsai "Diagnosable evaluation of general biswapped networks", in Processings of the 29th workshop on combinatorial mathematics and computation theory, pp. 272-276, 2012.

[3] K. Y. Chwa and S. L. Hakimi, "On fault identification in diagnosable systems," IEEE Trans. Computers, vol. C-30, pp. 414-422, 1981.

[4] J. Fan, "Diagnosability of the Möbius cubes", IEEE Trans. Parallel and Distributed Systems, vol. 40, no. 1, pp. 88-93, 1991.

[5] M. He, W. Xiao, W. Chen, W. Wei and Z. Zhang, "General Biswapped Networks and Their Topological Properties", Proc. of Advanced Parallel Processing Technologies (LNCS), vol. 4847, pp. 414-422, 2007.

[6] A. Kavianpour and A. D. Friedman, "Efficient design of easily diagnosable system," in: Proceedings of IEEE Computer Society's 3rd USA-Japan Computer Conference, 1978.

[7] A. Kavianpour and K. H. Kim, "Diagnosabilities of hypercubes under the pessimistic one-step diagnosis strategy," IEEE Trans. Computers, vol. 40, no. 2, pp. 232-237, 1991.

[8] A. Kavianpour, "Sequential diagnosability of star graphs," Computers Electrical Eng., vol. 22, no. 1, pp. 37-44, 1996.

[9] P. L. Lai, J. J. M. Tan, C. P. Chang, and L. H. Hsu, "Conditional Diagnosability Measures for Large Multiprocessor Systems," IEEE Transcations on Computers, vol. 54, no. 2, pp. 165-175, 2005.

[10] F. T. Leighton, Introduction to Parallel Algorithms and Architecture: Arrays, Trees, Hypercubes, Morgan Kaufmann, San Mateo, 1992.

[11] G. Marsden, P. Marchand, P. Harvey and S. Esener, "Optical transpose interconnection system architectures," Optics Letters, vol. 18, no. 13, pp. 1083-1085, 1993.

[12] F. P. Preparata, G. Metze, and R. T. Chien, "On the connection assignment problem of diagnosis systems", IEEE Trans. Electronic Computers, vol. 16, no. 12, pp. 848-854, 1967.

[13] C. H. Tsai, "A quick pessimistic diagnosis algorithm for hypercube-like multiprocessor systems under the PMC model," IEEE Transactions on Computers, vol. 62, no. 2, pp. 259-267, 2013.

[14] D. Wang, "Diagnosability of enhanced hypercubes," IEEE Trans. Computers, vol. 43, no. 9, pp. 1054-1061, 1994.

[15] C. H. Yeh and B. Parhami, "Swapped networks: unifying the architectures and algorithms of a wide class of hierarchical parallel processors," Proc. of Int. Conf. on Parallel and Distributed Systems, IEEE Computer Society, pp. 230-237, 1996.

[16] C. H. Yeh and B. Parhami, "Hierarchical swapped networks: efficient low-degree alternatives to hypercube and generalized hypercube," Proc. Int. Symp. on Parallel Architectures, Algorithms and Networks, IEEE Computer Society, pp. 90-96, 1996.

[17] C. H. Yeh and B. Parhami, "Recursive hierarchical swapped networks: versatile interconnection architectures for highly parallel systems," Proc. 8th IEEE Symp. on Parallel and Distributed Processing, IEEE Computer Society, pp. 453-460, 1996.